

CSA0711

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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I Nagarajan Subhash / 192312489 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Subhash
Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	2	2	2	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1/2	1/2	1/2	1/2	1/2

28
30

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Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Smart Campus Network Design

understanding the scenario

The smart campus has been allocated the IPv4 network 192.168.100.0/24. Since each department requires a different number of devices, variable length subnet masking (VLSM) is used to minimizing IP address wastage.

Application of Networking concepts

Department	Subnet	Network Address	Usable hosts
School of Computing	125	192.168.100.0	126
School of Electronics	126	192.168.100.128	62
School of Mechanical engineering	127	192.168.100.192	30

Remaining unused Address Range

192.168.100.224 - 192.168.100.255 (32 addresses)

Decision Making

The unused ip address should be reserved for future department or additional network devices. This subnetting plan improves scalability, reduces address wastage, and simplifies network management.

Conclusion :-

Variable length subnet/masking (VLSM) efficiently divides the available networks according to departmental requirements while ensuring future expansion.

Q. Multinational software company

Understanding the scenario

The company has been assigned 10.10.0.0/16, which provides a large address space. Different department requires different subnet sizes based on the number of employees.

Application of Networking concepts

Department	Subnet	Network Address	Broadcast Address	Usable hosts
Software Department	/24	10.10.0.0	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.127	126
Human-resources	/26	10.10.1.128	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.223	30

(24)

Remaining unused Address Range

10.10.1.224 - 10.10.255.255

Decision Making

Allocating subnet sizes according to department requirements ensures efficient bandwidth utilization reduces broadcast traffic, and leaves sufficient address for future traffic growth.

Conclusion :-

LSM provide an optimized and scalable network design while efficiently utilizing available IP address space.

IPv6 Migration

Understanding the scenario

The hospital is migrating from IPv4 to IPv6 to support thousands of medical devices and IoT sensors. IPv6 offers a significantly larger address space, simplified routing, and better scalability.

Given Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

compressed form

2001:db8:abcd::/64

expanded form

2001:0db8:abcd:0000:0000:0000:0000:0000

Network prefix

2001:db8:abcd:0000::/64

Host bits

bu bits

Total Host Addresses

$$2^{64} = 18,446,744, 073, 709, 551, 616$$

Decision making

IPv6 provides an enormous address space, making it ideal for smart hospitals with thousands of connected devices while improving network efficiency and future scalability.

Conclusion:-

Migrating to IPv6 eliminates address shortages and supports long-term expansion of healthcare networks.

routing

Decision

understanding the scenario

A router receives a packet with destination IP 192.168.2.45. The router compares the destination IP with the routing table and forwards the packet using the matching route.

Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Results :-

- destination subnet : 192.168.2.0/24
- matching routing entry : 192.168.2.0/24
- Next-Hop interface : interface connected to 192.168.2.0/24 network.

• Default route 0.0.0.0/0

Decision Making

The router forwards the packet using the matching subnet. if no match exists, it uses the default route to ensure packet delivery whenever possible.

Conclusion:-

Routing tables enable efficient packet forwarding and improve communication between interconnected networks.

Layer - 3 Router Configuration

Understanding the Scenario

Two departments are connected through a layer-3 router. Each department is assigned a separate subnet, and the router provides inter-network communication.

LAN	Network Address	Host Range
Administration	192.168.10.0/26	192.168.10.1 - 192.168.10.62
Finance	192.168.10.64/26	192.168.10.65 - 192.168.10.126

Suggested IP Address

Administration LAN

- PC1: 192.168.10.2
- PC2: 192.168.10.3
- PC3: 192.168.10.4

Finance LAN

- P11: 192.168.10.66
- P12: 192.168.10.67
- P13: 192.168.10.68

Default Gateways

- Administration: 192.168.10.1
- finance: 192.168.10.65

Decision Making

using separate subnets improves security and reduces broadcast traffic. The Layer-3 router enables communication between departments while maintaining efficient network performance.

Conclusion:-

Proper subnetting and gateway configuration provide reliable, secure, and scalable communication between departmental networks.